

SAI HARSHITHA AKULA

Generative AI Engineer | Applied AI Engineer | Machine Learning Engineer

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PROFESSIONAL SUMMARY

AI / Machine Learning Engineer with 4+ years of experience designing and deploying scalable Artificial Intelligence and Machine Learning solutions. Experienced in building Generative AI applications, Large Language Model (LLM) solutions, and Retrieval-Augmented Generation (RAG) pipelines using Python, LangChain, and modern ML frameworks. Skilled in developing end-to-end ML pipelines, semantic search systems, and AI-powered APIs while working with vector databases, NLP models, and cloud platforms (AWS, Azure, GCP). Proven ability to translate complex data into actionable insights and deliver production-ready AI systems that improve automation, decision-making, and user experience.

PROFESSIONAL EXPERIENCE

Dell Technologies, United States | *Applied AI Engineer*August 2024 – Present

- Designed and owned **end-to-end AI and Generative AI pipelines**, spanning **data ingestion, feature preparation, model fine-tuning, deployment, monitoring, and optimization** for enterprise use cases.
- Built and deployed **Retrieval-Augmented Generation (RAG) systems** using **LangChain, LlamaIndex, Pinecone, and Azure OpenAI**, improving factual response accuracy by **25%** and reducing document retrieval time by **35%**.
- Fine-tuned **embedding models and large language models (LLMs)** using **PyTorch and Hugging Face Transformers**, optimizing retrieval relevance and response grounding.
- Developed **agentic AI workflows** enabling multi-step reasoning, planning, and task automation to support complex enterprise workflows.
- Implemented **model evaluation and validation frameworks** (RAGAS, G-Eval) to systematically measure hallucination rate, precision, recall, and response relevance.
- Designed and deployed **scalable inference services** using **Docker and Azure Kubernetes Service (AKS)**, improving throughput and reducing latency by **30%**.
- Built and maintained **CI/CD pipelines for ML systems** using **GitHub Actions and MLflow**, supporting automated testing, model versioning, and governed releases.
- Implemented **monitoring and observability** using **Prometheus and Grafana** to track latency, error rates, and system health, maintaining **99.9% uptime** in production.
- Optimized **cloud resource utilization and inference workflows**, reducing operational costs by **20%** through batching, caching, and compute tuning.
- Partnered with **platform, security, and data engineering teams** to ensure AI systems met enterprise standards for **scalability, compliance, and security**.
- Supported internal stakeholders by translating business requirements into **reliable, production-ready AI solutions** with measurable impact.

Tiger Analytics, United States | *AI/ML Engineer*October 2023 – July 2024

- Contributed to **end-to-end ML pipelines** including data preprocessing, feature engineering, model training, evaluation, and deployment for analytics and automation use cases.
- Worked across **AWS and GCP client environments**, supporting ML workflows using **AWS SageMaker and GCP Vertex AI**.
- Built and tracked ML experiments using **MLflow**, improving retraining efficiency and experiment reproducibility.
- Developed **NLP and traditional ML models** using **scikit-learn, XGBoost, and transformer-based architectures**.
- Supported **semantic search and text summarization solutions** using embedding-based retrieval techniques.
- Integrated **OpenAI APIs** to enable natural-language interaction with analytics dashboards.
- Assisted with **model validation, retraining, and deployment automation** using CI/CD best practices.
- Implemented **SQL-based data quality checks and validation pipelines** to ensure reliable model inputs.
- Collaborated with cross-functional teams to translate business requirements into deployable ML solutions.

WNS Global Services, India | *ML Engineer*July 2021 – July 2023

- Built **predictive ML models** for **sales forecasting** and **customer segmentation** using **Python, scikit-learn, and SQL**.
- Owned the **full ML lifecycle** including **feature engineering, model selection, hyperparameter tuning, model evaluation, deployment, and retraining workflows**.
- Developed **ETL pipelines** and **data preprocessing workflows** using **Pandas, NumPy, and SQL** to ensure **clean, reliable training data**.
- Automated **ML pipelines** using **cloud-based orchestration and scheduling tools**, reducing **manual retraining time** by **35%**.
- Deployed **ML models** as **REST APIs** using **Flask-based services** on **Google Cloud** for **real-time inference**.
- Built **Tableau dashboards** and **Matplotlib visualizations** to track **KPIs, forecasts, and model performance**.
- Delivered **data-driven insights** that improved **forecasting accuracy** and optimized **operational decision-making**.

PROJECTS

Enterprise Knowledge Assistant using RAG

- Built an enterprise knowledge assistant using Retrieval-Augmented Generation (RAG) to enable semantic search across large document repositories.
- Implemented document ingestion pipelines with chunking, embeddings, and vector indexing using FAISS.
- Integrated LLM-based contextual answering using OpenAI APIs and LangChain to improve response relevance.
- Exposed the system through FastAPI REST endpoints, enabling integration with internal enterprise applications.

Multi-Agent AI Workflow Automation System

- Designed a multi-agent AI system to automate multi-step reasoning tasks using LangGraph and LangChain.
- Implemented tool-calling agents capable of retrieving information, summarizing data, and executing workflows.
- Built modular agents for data retrieval, reasoning, and response generation, improving workflow automation efficiency.
- Developed evaluation pipelines to measure response accuracy, reasoning steps, and tool usage performance.

Intelligent Document Processing System (OCR + NLP)

- Developed an AI-powered document processing pipeline to extract structured insights from unstructured documents.
- Integrated OCR pipelines for document digitization and applied NLP models for entity extraction and classification.
- Built ML models to automate document categorization and metadata tagging.
- Deployed the solution using cloud-based ML services, enabling scalable document processing.

AI-Powered Conversational Analytics Chatbot

- Developed a conversational AI assistant capable of answering analytics questions using natural language queries.
- Implemented vector-based semantic search for contextual retrieval of relevant data insights.
- Integrated LLM-based summarization and explanation capabilities to improve decision support.
- Built a Streamlit-based interactive interface for real-time conversational analytics.

TECHNICAL SKILLS

Languages:

Python, SQL, R

Machine Learning & Deep Learning:

PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM

Generative AI & LLMs:

LLMs (GPT family, Claude, LLaMA), RAG, LangChain, LlamaIndex, Hugging Face Transformers, OpenAI APIs

Model Training & Evaluation:

Feature Engineering, Hyperparameter Tuning, Cross-Validation, Evaluation Metrics (Accuracy, F1, Precision/Recall, AUC), Error Analysis

Cloud Platforms:

Microsoft Azure (Azure OpenAI, Cognitive Search, AKS), AWS (SageMaker, Lambda), Google Cloud (Vertex AI, BigQuery)

MLOps & Deployment:

MLflow, Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins), Model Serving (FastAPI/Flask), Model Versioning

Monitoring & Observability:

Prometheus, Grafana, CloudWatch, Data Drift Detection

Data Engineering & ETL:

Pandas, NumPy, SQL-based ETL, Apache Airflow (basic exposure), AWS Glue

Visualization & Business Intelligence:

Power BI, Tableau, Streamlit, Jupyter Notebooks

EDUCATION

Kent State University, OHIO

Master of Science in Business Analytics

May 2025